

RICHAGO-4G

**Omega-3 Fatty Acids + Green Tea Extract + Ginseng + Ginkgo Biloba + Multivitamins +
Multiminerals + Trace Elements
Softgel Capsules**

1. Product Overview

RICHAGO-4G is a comprehensive nutraceutical formulation combining essential fatty acids, botanical antioxidant extracts, vitamins, minerals, and trace elements intended to support metabolic function, cellular defense, cognitive performance, cardiovascular wellness, and recovery during periods of physiological stress or nutritional insufficiency.

The formulation integrates adaptogenic, antioxidant, neuroprotective, and micronutrient-supportive components designed to address oxidative stress, fatigue, reduced vitality, and impaired immune resilience.

2. Component Synergy

The formulation utilizes a multi-component antioxidant and metabolic support approach in which botanical extracts and micronutrients act synergistically to reduce oxidative burden and support cellular homeostasis.

Ginseng

- Rich in ginsenosides with adaptogenic and immunomodulatory properties
- Supports hypothalamic-pituitary-adrenal (HPA) axis regulation during stress
- Enhances cellular energy metabolism and physical endurance
- Demonstrates antioxidant activity through reduction of reactive oxygen species (ROS)

Ginkgo Biloba

- Contains flavone glycosides and terpene lactones with antioxidant and microcirculatory effects

- Supports cerebral and peripheral blood flow
- Helps reduce oxidative neuronal injury
- May assist in maintenance of cognitive and vascular function

Green Tea Extract

- Standardized polyphenols, particularly epigallocatechin gallate (EGCG), provide potent free radical scavenging activity
- Helps modulate inflammatory pathways and oxidative stress
- Supports endothelial function and metabolic health
- Exhibits protective effects against lipid peroxidation

Grapeseed-Derived Polyphenolic Antioxidant Activity (“4G Concept”)

The combined antioxidant network generated by:

- Ginseng
- Ginkgo Biloba
- Green Tea polyphenols
- Polyphenolic antioxidant support

contributes to broad-spectrum cellular protection against oxidative damage affecting vascular, neural, and immune tissues.

Omega-3 Fatty Acids

- Provide eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA)
- Support membrane fluidity and anti-inflammatory mediator production
- Contribute to cardiovascular and neurological health
- Help regulate inflammatory cytokine responses

Multivitamins, Multiminerals, and Trace Elements

- Support enzymatic reactions essential for energy generation and immune competence
 - Assist mitochondrial metabolism and antioxidant enzyme function
 - Help correct subclinical micronutrient deficiencies associated with fatigue and stress
-

3. Therapeutic Rationale

RICHAGO-4G may be used as nutritional support in conditions associated with increased metabolic demand, oxidative stress, or nutritional insufficiency.

Convalescence and Recovery

- Supports tissue repair and metabolic recovery following illness, surgery, or prolonged physiological stress
- Assists restoration of nutritional reserves and cellular function

Fatigue and Reduced Vitality

- Adaptogenic botanical components may improve stress tolerance and perceived energy levels
- B-complex vitamins and trace minerals support mitochondrial ATP production and neuromuscular function

Immune Support

- Antioxidants and micronutrients contribute to normal immune cell activity
- Oxidative stress modulation may support immune resilience during chronic stress states

Cognitive and Vascular Support

- Ginkgo Biloba and Omega-3 fatty acids may support cerebral circulation and neuronal membrane integrity
- Useful in individuals with age-related nutritional and oxidative stress concerns

Nutritional Supplementation

- May be considered in dietary insufficiency, inadequate intake, or increased nutritional demand

4. Scientific Evidence

Adaptogenic Effects of Ginseng

Clinical studies evaluating standardized Panax ginseng preparations have reported:

- Improvement in fatigue perception
- Enhanced physical endurance and mental performance
- Better stress adaptation through modulation of neuroendocrine pathways

Experimental evidence suggests ginsenosides influence inflammatory mediators, oxidative stress pathways, and neurotransmitter regulation associated with fatigue and cognitive performance.

Cardiovascular Benefits of Omega-3 Fatty Acids

Large-scale clinical and epidemiological studies have demonstrated that Omega-3 fatty acids:

- Support normal triglyceride metabolism
- Contribute to endothelial function
- Exhibit anti-inflammatory cardiovascular effects
- May help maintain normal cardiac electrophysiological stability

EPA and DHA have also been associated with modulation of vascular inflammation and maintenance of healthy lipid profiles.

Antioxidant and Neurovascular Support

Polyphenolic compounds from Green Tea and Ginkgo Biloba have demonstrated:

- Reduction in oxidative stress biomarkers
 - Endothelial protective effects
 - Supportive actions on cerebral microcirculation and cellular antioxidant defense systems
-

5. Safety Profile and Precautions

Ginkgo Biloba and Anticoagulants

Use with caution in patients receiving:

- Warfarin

- Aspirin
- Clopidogrel
- Other antiplatelet or anticoagulant therapies

Ginkgo may theoretically increase bleeding tendency due to platelet-activating factor modulation.

Omega-3 Fatty Acids

High intake may contribute to prolonged bleeding time in susceptible individuals or in combination with anticoagulants.

Caffeine Sensitivity

Green Tea Extract may contain naturally occurring caffeine, which can contribute to:

- Insomnia
- Nervousness
- Palpitations
- Gastric discomfort in sensitive individuals

Hypersensitivity

Avoid use in individuals with known hypersensitivity to any formulation component.

Pregnancy and Lactation

Use during pregnancy and lactation should be based on physician recommendation.

General Precautions

Use cautiously in:

- Patients with bleeding disorders
- Individuals with severe hepatic dysfunction
- Patients scheduled for surgery

6. Adverse Effects

Adverse effects are generally mild and infrequent when used at recommended dosages.

Possible Reactions

- Gastrointestinal discomfort
 - Nausea
 - Mild dyspepsia
 - Headache
 - Insomnia or restlessness
 - Allergic reactions (rare)
-

7. Dosage and Administration

Recommended Daily Intake

- One softgel capsule once daily after meals
- Or as directed by the healthcare professional

Method of Administration

- Swallow whole with water
 - Preferably administered after food to improve tolerability
-

8. Storage

- Store below 25°C
 - Protect from heat, moisture, and direct sunlight
 - Keep out of reach of children
-

9. Dosage Form

Softgel Capsules for oral administration.

